

Project Title: VoidView: AI-Powered Retail Location Intelligence & Viability Prediction System

Domain: Machine Learning / Geospatial Data Science / Location Intelligence

Abstract / Synopsis:

Selecting the right brick-and-mortar location is the single most critical factor determining the survival of a retail business. Traditional site selection relies heavily on subjective intuition or generic commercial footfall data, leading to high retail closure rates due to hidden competitor saturation or poor demographic alignment.

VoidView ("*From void to viability*") is an AI-driven geospatial decision support system designed to evaluate and predict retail site viability across Kalyan City.

Key Methodologies & Features:

1. **Data Collection & Feature Engineering:** A dataset of 267 real-world commercial entities was gathered across 10 retail categories using the Google Places API. Features include competitor density within 500m, nearest competitor proximity, total commercial density within 1km, average ratings, review volumes, and one-hot encoded business categories.
2. **Machine Learning Model:** A balanced Logistic Regression classifier was trained with StandardScaler normalization and 5-fold cross-validation, achieving an ROC-AUC score of **0.8459** for business survival/closure prediction.
3. **Spatial Intelligence Layer:** The system integrates a dynamic **Pedestrian Footfall Index (0–100)** derived from distance-weighted transit hubs, an **Anchor Tenant Synergy Engine** modeling positive footfall spillovers from 13 major landmarks (hospitals, colleges, transit terminals), and a **Demographic Zone Multiplier** covering key Kalyan neighborhoods.
4. **Interactive Deployment:** VoidView is deployed as a zero-latency, serverless client-side web application using Leaflet.js and hosted on GitHub Pages. Users can click any coordinate on the interactive map or search local landmarks to obtain an instant viability score (0–100%), full factor impact explanations, and strategic location advice.